The In-Ga-Sb association of the post-Variscan Zn-Pb-Ag vein system at Lautenthal, Upper Harz Mountains, Germany: sphalerite mineral chemistry.

Graupner T, Henning S, Goldmann S. Fuchs S, Stedingk K, Liessmann W, Birkenfeld S.

Mineralium Deposita.

Electronic Supplementary Material (ESM)

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1. Description of methods used (extended)

Energy-dispersive X-ray fluorescence spectrometry mapping (μ-EDXRF) at BGR (Hannover):

The X-ray beam of the Rh-tube was focused on the sample surface with an angle of 51° and a spot size diameter of approximately 17 μm at 17.5 keV. Mapping of the samples with maximum excitation (tube: 50 kV, 600 μA), 2 ms measuring time per spot, and step ranges between 20 to 50 μm yielded spatially resolved analytical data, with elements heavier than Na measurable under low vacuum conditions (Flude et al. 2017).

In-situ laser ablation-inductively coupled plasma-mass spectrometry (LA-ICP-MS) at BGR (Hannover):

In the analytical sessions, the sphalerite reference material MUL-ZnS-1 (Onuk et al. 2017) and the polymetallic reference material MASS-01 (Danyushevsky et al. 2011) were ablated several times. Element concentrations in sphalerite were quantified using MUL-ZnS-1 as calibration material and MASS-01 as reference material for quality control (Wilson et al. 2002; Yuan et al. 2012). Element concentrations in chalcopyrite were quantified using MASS-01 as calibration material. Raw data were exported and processed with a data-handling software tool (Gäbler et al. 2011). A calibration strategy based on ablation yield correction factors (Liu et al. 2008) was used for internal standardization, in which the sum of all cations was normalized to the stoichiometric content of Zn in sphalerite (67.0 wt%) and the stoichiometric content of Cu and Fe in chalcopyrite (65.1 wt%). Chalcopyrite was analyzed using 107 LA-ICP-MS spot measurements in 14 samples.

Electron probe microanalysis (EPMA) at BGR (Hannover):

Major elements were analyzed for 8-40 s applying sphalerite as reference material for Zn and S, chalcopyrite for Cu and siderite (carbonates) and pyrite (sulfides) for Fe using their respective Kα X-ray lines. Minor and trace element signals were counted for 60-120 s using a PET crystal for In Lα and a LIF crystal for Ge Kα and Ga Kα. For calibration, GaAs as well as pure metals (In and Ge) were used as reference materials.

Quantitative EPMA single spot datasets confirm the major and minor element compositions of sphalerite as determined by LA-ICP-MS analysis within identical areas of the grains, but EPMA data are often close to or below the LOD of the method for some trace elements.

Determination of the base metal chemistry and trace element composition of sphalerite grain separates and laboratory ore concentrates:

Sphalerite grain separates (sample type A): Precision and accuracy are ensured by blank runs, in-house control, duplicate analysis and a suite of internationally certified standards by the laboratories of Actlabs (Activation Laboratories Ltd., Canada).

Laboratory ore concentrates (sample type B, first batch) were crushed and subsequently milled using a vibrating cup mill for the duration of 15 seconds (TU Clausthal, Clausthal-Zellerfeld, Germany). The resulting material was initially digested in a matrix of aqua regia, HF and then complexed with H_3BO_3 prior to analysis. Chemical analysis was carried out using an ICP-OES (VISTA MPX by Varian). The limits of detection for Ga and In were both at about 2.0 ppm. The second batch of Type B concentrates was reduced in size by jaw crushers, a roller mill and a laboratory drum mill. Following aqua regia digestion of the material, the major elements (Cu, Fe, Zn, Pb) and Ga were analyzed by ICP-OES (ICP-OES 5100; Agilent) and the trace element In by ICP-MS (ICP-MS iCAP Q; Thermo Fisher Scientific). For further details of the methods used for crushing and chemical analysis of the second batch of Type B concentrates the reader is referred to Haas (2020).

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Figure ESM 1 Correlation trends for selected trace elements in In-bearing sphalerite-2b of the carbonate-sulfide stage (LA-ICP-MS data) from Lautenthal (Bromberg orebody). BSG - Bergstern vein (370 m West; 2 samples); BEM - Lautenthal vein (700 m West; 10 samples).

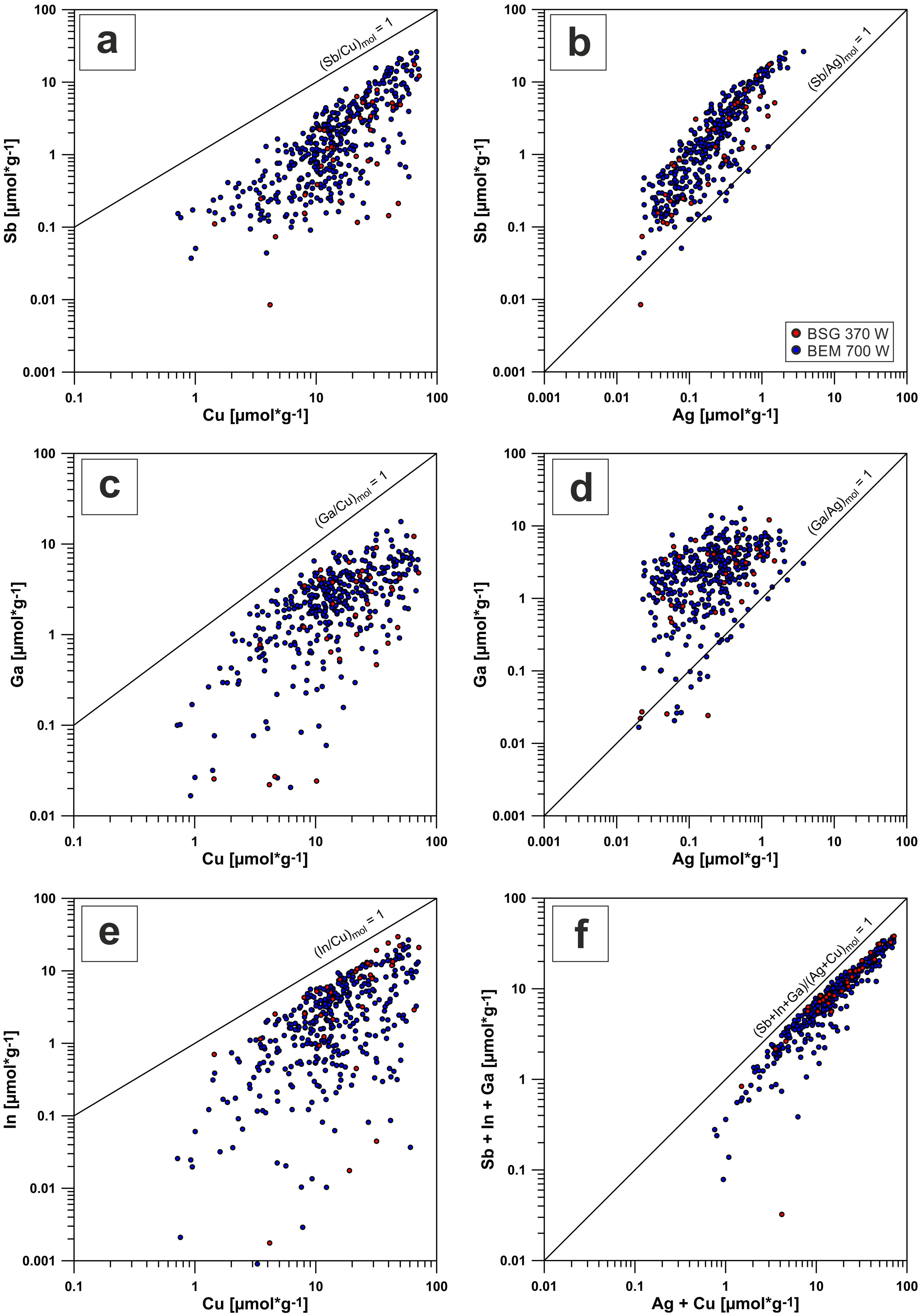

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Table ESM_1 Studied locations and numbers of samples taken for both orebodies of the Lautenthal deposit

Orebody	Mine	Vein	Mining level	Location	Ore stage(s)	Number of samples	Hand-picked sphalerite samples (weight (g))	Ore concen-trates (0.5 to >50 kg each)	Origin of samples
Bromberg orebody	Lautenthal's Glück mine	Lautenthal vein	Ernst August gallery level (+200 mMSL)	Western exploration drift, Crosscut 500 m West	quartz-sulfide	5	3 (48, 57, 68)	1 + 1*	Field work 2017
				Western exploration drift, Crosscut 600 m West	quartz-sulfide	2	1 (51)		Field work 2018
				Western exploration drift, Crosscut 700 m West, footwall branch vein	quartz-sulfide	3	2 (56, 58)	2	Field work 2017
				Western exploration drift, Crosscut 700 m West, central branch vein	quartz-sulfide / carbonate-sulfide	13	2 (69, 71)	2 + 1*	Field work 2017, 2018
		Bergstern vein	Ernst August gallery level (+200 mMSL)	Western exploration drift, Outcrop 330 m West	quartz-sulfide	1			Field work 2014
				Western exploration drift, Crosscut 370 m West	carbonate-sulfide	4			Field work 2014
Lautenthal orebody	Güte des Herrn mine	Lautenthal vein	12^{th} level (-20 mMSL)	12^{th} Feldortstreckenfirste, Lautenthals' Glück branch vein	quartz-sulfide	3			Ore deposit collection (TU Clausthal)
			8^{th} level (+90 mMSL)	8^{th} Feldortstrecke, 203 m East of shaft, Lauten-thals' Glück branch vein	quartz-sulfide / carbonate-sulfide	1			Ore deposit collection (BGR)
	Lautenthal's Glück mine	Lautenthal vein	Ernst August gallery level (+200 mMSL)	Outcrop close to crosscut 100 m East, Lautenthal's Glück branch vein, „Kalkspatfirste“	quartz-sulfide / carbonate-sulfide	2			Field work 2014, 2015
	Güte des Herrn mine		ca. +200 mMSL	Abendstern branch vein	quartz-sulfide / carbonate-sulfide	1			Ore deposit collection (TU Clausthal)
	Schwarze Grube mine (Black mine)	Jakob vein	+120 mMSL	Lower „Feldortstrecke“, 230 m East of the shaft	quartz-sulfide	1			Ore deposit collection (BGR)

* - preparation of ore concentrates from the large volume samples and chemical analysis of ore concentrates by A. Haas (2020)

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ESM

Table ESM_2 Major, minor and trace element composition of sphalerite from the Bromberg and Lautenthal orebodies of the hydrothermal vein deposit at Lautenthal. All samples were analyzed using the Laser ablation-ICP-MS method, unless otherwise stated.

Orebody / vein / location and ore type	Ore stage / ore sub-stage	Number of samples/ measurements (n)	Fe (wt%)			Cd (wt%)		Mn (ppm)		Cu (ppm)		Sb (ppm)		Sn (ppm)		Ag (ppm)		Ga (ppm)		Ge (ppm)		In (ppm)		GGIMFis temperature***
			min-max	mean	median	min-max	median	min-max	median	min-max	median	min-max	median	min-max	median	min-max	median	min-max	median	min-max	median	min-max	median	median for samples (°C)
Bromberg orebody																								
Lautenthal vein																								
Lautenthal's Glück mine, Ernst August gallery level (+200 mMSL)																								
Crosscut 500 m West; quartz-sphalerite-1 banded and breccia ore	quartz-sulfide stage	5 / 217	0.97-2.94	1.96	1.97	0.11-0.54	0.21	10.8-55.1(184)	24.7	<LOD-2600	158	<LOD-1300	42.2	<LOD-43.4	<LOD	1.00-208	15.0	<LOD-463	36.5	<LOD-180	8.62	<LOD-29.8	<LOD	156; 157; 174; 191; 204
Crosscut 600 m West; quartz-sphalerite-1 banded and breccia ore	quartz-sulfide stage	2 / 80	0.44-2.57	1.64	1.68	0.11-0.91	0.32	2.94-44.1	21.3	13.2-6060	259	0.25-2620	147	<LOD-1.67	<LOD	0.48-217	18.2	<LOD-147	27.9	<LOD-55.5	5.84	<LOD-10.9	0.12	179; 180
Crosscut 700 m West, footwall branch vein; quartz-sphalerite-1 banded and breccia ore	quartz-sulfide stage	3 / 155	0.37-2.88	1.38	1.19	0.14-0.86	0.26	<LOD-42.6	9.30	14.3-3720	301	0.49-2020	96.0	<LOD-20.5	<LOD	1.51-175	15.4	<LOD-655	65.6	<LOD-189	10.6	<LOD-744	1.28	151; 166; 175
Crosscut 700 m West, central branch vein; quartz-sphalerite-1 banded (massive) and breccia ore	quartz-sulfide stage	1 / 35	1.15-2.12	1.69	1.70	0.14-0.71	0.25	<LOD-15.0	10.3	12.5-4010	279	<LOD-1900	80.8	<LOD-19.5	<LOD	0.81-152	12.2	<LOD-511	104	<LOD-189	10.7	<LOD-84.6	1.18	165
Crosscut 700 m West, central branch vein; quartz-galena-2a and quartz-sphalerite-2a stringers in footwall part of branch vein	carbonate-sulfide stage / quartz-dominated sub-stage	1 / 35	0.75-2.12	1.20	1.11	0.18-0.78	0.28	<LOD-16.5	8.70	7.27-3400	348	<LOD-1820	48.3	<LOD-148	<LOD	1.02-142	6.65	<LOD-579	134	<LOD-206	11.0	<LOD-952	13.2	163
Crosscut 700 m West, central branch vein, vein center; calcite-sphalerite-2b-galena ore	carbonate-sulfide stage / calcite-dominated sub-stage	10 / 392	0.29-2.32	1.20	1.13	0.13-0.83	0.26	<LOD-52.5 (2870)	8.24	28.4-4460	830	<LOD-3200	156	<LOD-467	11.8	0.52-404	20.3	<LOD-1233	165	<LOD-182	9.31	<LOD-3060	238	158; 166; 174; 182; 184; 184; 187; 187; 188; 210
Bergstern vein																								
Lautenthal's Glück mine, Ernst August gallery level (+200 mMSL)																								
Outcrop 330 m West; quartz-sphalerite-1 breccia	quartz-sulfide stage	1 / 30	0.96-2.27	1.54	1.57	0.15-0.61	0.25	14.8-37.0	24.7	4.18-2370	80.4	<LOD-796	7.86	<LOD-217	<LOD	0.99-62.1	2.24	<LOD-579	53.3	<LOD-126	6.81	<LOD-155 (1420)	0.71	189
Crosscut 370 m West; quartz-sphalerite breccia ore	carbonate-sulfide stage (or quartz-sulfide stage)	2 / 83	0.43-1.73	1.11	1.06	0.13-0.46	0.25	1.58-17.9	10.5	8.94-3740	478	0.38-1920	215	<LOD-64.6	1.01	0.43-160	35.7	0.63-806	132	<LOD-122	11.1	<LOD-953	4.76	160
Crosscut 370 m West; galena-2-rich, sphalerite-2b-bearing veinlets cutting ore breccia	carbonate-sulfide stage / calcite-quartz-dominated sub-stage	2 / 37	0.13-2.40	0.91	0.91	0.13-0.74	0.25	<LOD-14.5	7.12	91.8-4530 (13300)	1283	1.03-2130	115	<LOD-372	31.7	1.71-162	23.5	0.69-845	153	<LOD-160	7.01	0.20-3390	574	174; 200
Lautenthal orebody																								
Lautenthal vein																								
Lautenthal's Glück branch vein																								
Güte des Herrn mine, 12th level (-20 mMSL)																								
Samples from ore deposit collection; massive or banded quartz-(calcite)-sphalerite-1-galena ore	quartz-sulfide stage	3/190	0.43-4.09	2.28	2.28	0.11-0.68	0.19	<LOD-277	35.1	<LOD-1220	144	<LOD-610	21.9	<LOD-328	<LOD	1.00-131	11.4	<LOD-354	15.0	<LOD-163	1.02	<LOD-414	0.76	202; 241; 246
Güte des Herrn mine, 8th level (+90 mMSL)																								
Sample from ore deposit collection, 203 m East of the shaft; quartz-sphalerite-1-galena breccia ore; sphalerite-1 partly replaced by chalcopyrite	quartz-sulfide stage overprinted by carbonate-sulfide stage fluids	1 / 30	2.03-3.90	3.16	3.22	0.18-0.53	0.28	37.5-110	77.7	93.9-2680	802	0.77-818	35.7	<LOD-460	6.91	3.84-48.6	15.1	<LOD-664	85.4	<LOD-89.0	6.23	0.16-671	10.6	226
Lautenthal's Glück mine, Ernst August gallery level (+200 mMSL)																								
Outcrop close to crosscut 100 m East, "Kalkspatfirste"; deformed sphalerite-1 fragments partly replaced by chalcopyrite	quartz-sulfide stage overprinted by carbonate-sulfide stage fluids	2 / 41	0.72-3.17	2.24	2.32	0.17-0.61	0.29	7.12-96.7	51.8	44.3-3470 (12899)	1124	0.45-986	17.9	<LOD-685	6.30	1.81-62.8	16.9	<LOD-381	24.7	<LOD-161	5.90	<LOD-1230	15.6	198; 244
Abendstern branch vein																								
Güte des Herrn mine, Ernst August gallery level (+200 mMSL)																								
Sample from ore deposit collection; quartz-(calcite), sphalerite-1 is replaced by chalcopyrite to large parts	quartz-sulfide stage overprinted by carbonate-sulfide stage fluids	1 / 20*	1.51-2.94*	2.22*/2.05**	2.08*	0.20-0.63*	0.27*		124**	686-4200*	2310*	<LOD-711*	64.0**	<LOD-1120*	32.3**	227-549*	32.8**	<LOD-401*	64.7**	<LOD-116*	20.1**	<LOD-2330*	82.6**	
Jakob vein																								
Schwarze Grube - Black mine (+120 mMSL)																								
Sample from ore deposit collection, 230 m East of the shaft; quartz-(calcite)-sphalerite-1 breccia ore	quartz-sulfide stage	1 / 40	1.39-3.00	1.99	1.95	0.14-0.43	0.19	31.3-74.8	43.6	25.7-1399	430	<LOD-323	9.33	<LOD-318	2.45	2.09-35.3	9.66	<LOD-459	6.08	<LOD-154	<LOD	<LOD-578	10.2	244
Average Detection Limits of LA-ICP-MS (LOD)				0.1 wt%		2.0 ppm		5.0 ppm		3.0 ppm		0.4 ppm		1.0 ppm		0.4 ppm		1.0 ppm		2.0 ppm		0.1 ppm		

Legend: * - data are from EPMA; ** - mean of data for about 50.000 pixels (LA-ICP-TOF-MS data); *** - calculated crystallization temperatures of sphalerite using the geothermometer GGIMFis (Frenzel et al. 2016); values below LOD were set to 50% of LOD for calculation of GGIMFis temperatures